

Lab 1a – Bash CLI Runbook (Verified Working)

This document provides the step-by-step Bash / AWS CLI process used to successfully deploy Lab 1a with EC2, RDS, IAM, Secrets Manager, and SSM in us-east-1.

Prerequisites

- AWS CLI installed and authenticated
- Region set to us-east-1
- IAM permissions for EC2, RDS, IAM, Secrets Manager, SSM

VPC & Networking

- VPC CIDR: 10.20.0.0/16
- Public subnet for EC2
- Two private subnets in different AZs for RDS
- Internet Gateway attached
- Route table with 0.0.0.0/0 to IGW

Security Groups

- EC2 SG: allow 22, 80, 5000 inbound
- RDS SG: allow 3306 ONLY from EC2 SG

RDS Deployment

- Create DB subnet group (private subnets)
- Launch MySQL RDS (not public)
- Capture endpoint and port

Secrets Manager

- Create secret: lab/rds/mysql
- Store DB credentials and endpoint

IAM Configuration

- Create EC2 IAM role
- Attach AmazonSSMManagedInstanceCore
- Attach inline Secrets Manager read policy
- Create instance profile

EC2 Launch

- Launch Amazon Linux 2023 instance
- Attach instance profile
- Provide user-data script
- Verify instance is running

Application Verification

- `curl http://EC2_PUBLIC_IP/`
- `curl http://EC2_PUBLIC_IP/init`
- `curl http://EC2_PUBLIC_IP/add?note=test`
- `curl http://EC2_PUBLIC_IP/list`

Success Criteria

- HTTP 200 responses
- Notes persist in RDS
- SSM access works
- No public RDS access